

AI researcher specializing in advanced reasoning, multimodal foundation models, autonomous agents, and AI for science. I am a researcher at Shanghai AI Lab. I received my Ph.D. in Machine Learning and Intelligence from Peking University in 2025. I have hands-on experience contributing to trillion-scale foundation models, reinforcement learning systems, and advanced reasoning agents through research roles at Shanghai AI Lab, Tencent, ByteDance, and Microsoft Research Asia. My research develops scalable learning methods for mathematical reasoning, scientific discovery, and generalizable intelligent agents, with publications at ICLR, NeurIPS, ICML, ACL, COLM, CVPR, among others.

EDUCATION

Ph.D. in Machine Learning and Intelligence, Peking University, advised by Zhihong Deng	September 2020 – July 2025
Bachelor of Science in Psychology, Peking University	September 2016 – July 2020

SELECTED RESEARCH

Advanced Reasoning & AI for Mathematics

- Olympiad-Level Mathematical Agents** Developed informal and formal mathematical agents, including Intern-S1-MO and InternGeometry. The system scored 102 in the official 2025 CMO AI track, reaching the Gold Medal threshold and ranking third under the same evaluation conditions as human contestants
Selected work: Achieving Olympia-Level Geometry Large Language Model Agent via Complexity Boosting Reinforcement Learning, ICLR 2026; Long-horizon Reasoning Agent for Olympiad-Level Mathematical Problem Solving, COLM 2026
- Research-Level Reasoning and Self-Evolution** Proposed scaffolding-internalization training to transfer the capabilities of advanced reasoning agents back into base models, and developed agents that iteratively collaborated with human experts on research-level mathematics
Selected work: Two working papers

Foundation Models & Multimodal Pretraining

- Trillion-Scale Scientific Foundation Models** Contributed advanced mathematical reasoning and multimodal capabilities to the Intern-S model family, including Intern-S1, Intern-S1-Pro, and S2-Preview
Selected work: Intern-S1-Pro: Scientific Multimodal Foundation Model at Trillion Scale, arXiv; Intern-S1: A Scientific Multimodal Foundation Model, arXiv
- Scalable Visual Pretraining for Language Intelligence** Developed next-patch prediction methods for visual pretraining language models on large-scale PDF corpora, demonstrating that pure visual pretraining can improve textual language capabilities
Selected work: Exploring Visual Pretraining for Learning Language Intelligence, CVPR 2026; Scalable Visual Pretraining for Language Intelligence, arXiv

Autonomous Agents & AI for Science

- Agent Skill Learning and Evaluation** Developed methods that enable language-model agents to create, refine, and reuse action skills during task execution, and contributed to evaluations of multimodal agents in realistic scientific workflows
Selected work: Empowering Large Language Model Agents through Action Learning, COLM 2024; ScienceBoard: Evaluating Multimodal Autonomous Agents in Realistic Scientific Workflows, ICLR 2026
- Multimodal Learning for Biology and Chemistry** Developed unified graph-text models for molecular generation, zero-shot molecular learning, protein representation learning, and biological reasoning
Selected work: GIMLET: A Unified Graph-Text Model for Instruction-Based Molecule Zero-Shot Learning, NeurIPS 2023; Retrieved Sequence Augmentation for Protein Representation Learning, EMNLP 2024; Are More Layers Beneficial to Graph Transformers?, ICLR 2023

Publication record: 20+ papers and preprints, including publications at ICLR, NeurIPS, ICML, ACL, EMNLP, COLM, and CVPR. Complete list: [Personal Website](#) / [Google Scholar](#).

EXPERIENCE

Full-time Researcher LLM Lab	July 2025 – Present Shanghai AI Lab
Research Intern LLM Lab	October 2024 – July 2025 Shanghai AI Lab
Research Intern AI Lab NLP	March 2024 – October 2024 Tencent

Research Intern
Data-AML-US, mentored by Jingjing Xu

August 2023 – February 2024
ByteDance

Research Intern
HKU NLP, advised by Lingpeng Kong and Qi Liu

September 2022 – July 2025
The University of Hong Kong

Research Intern
Natural Language Computing Group, mentored by Shuming Ma

January 2022 – October 2022
Microsoft Research Asia

Collaborator
NAIL Group, advised by Luu Anh Tuan

June 2021 – July 2025
Nanyang Technological University

Research Assistant
Sigma Lab, Department of Machine Intelligence

July 2018 – July 2020
Peking University

AWARDS & HONORS

2022 Guorui Scholarship, Peking University
2022 Merit Student Award, Peking University
2018 Scientific Research Award, Peking University